1B Multiplying and Dividing Integers

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Evaluate each integer product:

(a) -7×8 (b) $-9 \times (-12)$ (c) $15 \times (-6)$ (d) $-4 \times (-5) \times (-6)$

Q2. Evaluate each integer quotient:

(a) $-72 \div 9$ (b) $-96 \div (-12)$ (c) $108 \div (-9)$ (d) $\dfrac{-144}{16}$

Q3. Evaluate each power, paying careful attention to brackets:

(a) $(-6)^2$ (b) -6^2 (c) $(-4)^3$ (d) $(-2)^4 \times (-3)^2$

Q4. Evaluate each combined multiplication and division expression:

(a) $-48 \div (-6) \times (-3)$ (b) $\dfrac{-15 \times (-8)}{-10}$ (c) $(-3)^3 \div 9 \times (-5)$ (d) $\dfrac{(-12) \times 14}{-21}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A hot-air balloon descending over the Yarra Valley loses altitude at a constant rate of $-18\ \text{m}$ per minute. (a) What is its change in altitude over 7 minutes? (b) How many minutes does it take for its altitude to change by $-234\ \text{m}$?

Q6. During a dry summer week in Mildura, an irrigation dam's water volume changed by $-420\ \text{kL}$ over 6 equal days. What was the mean daily change in volume?

Q7. In a Year 8 scholarship quiz of 25 questions, a correct answer scores $+5$ points, an incorrect answer scores -3 points, and an unanswered question scores -1 point. Liam answers 17 correctly, gets 5 incorrect, and leaves 3 unanswered. Calculate his total score.

Q8. The minimum overnight temperatures in Ballarat over 5 consecutive winter nights were -3°C, -5°C, -1°C, -4°C, and -2°C. Calculate the mean minimum temperature across the 5 nights.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Find all pairs of integers (x, y) with $x < y$ such that $x \times y = -24$ and $x + y = -5$. Show how you tested the factor pairs of -24.

Q10. Without calculating the full numerical value, determine whether the product $(-1)^1 \times (-2)^2 \times (-3)^3 \times (-4)^4 \times (-5)^5$ is positive or negative. Explain your reasoning by counting negative factors.